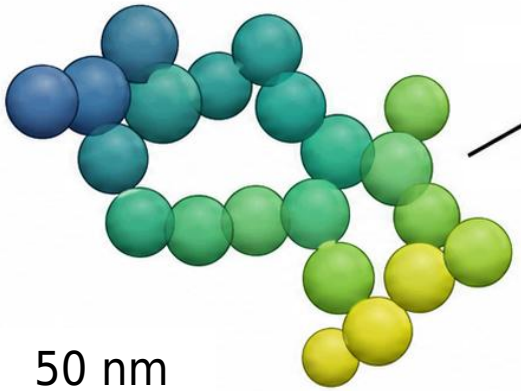
Aggregate



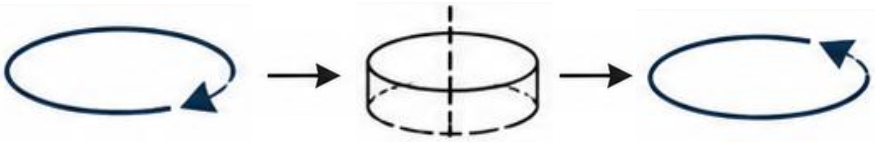
$\{\boldsymbol{r}_i, a_i, m_i\}$

sphere T matrices T_i

RCR T-matrix construction

RCR translation between sphere centers

$$\mathbf{U}_{ij} = \mathbf{D}^{-1}(\boldsymbol{r}_{ij})\,\mathbf{C}(\boldsymbol{r}_{ij})\,\mathbf{D}(\boldsymbol{r}_{ij})$$



rotate coaxial rotate back

multi-RHS coupled solve

$$\mathbf{AX} = \mathbf{B}_{\text{rhs}}$$

aggregate T matrix

$$\mathbf{T}_{\text{agg}} = \mathbf{M}_{\text{flat}}\mathbf{X}\mathbf{P}_{\text{col}}$$

Observables

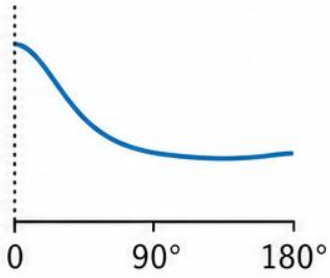
Q_{ext}, Q_{sca}, Q_{abs}

SSA, g

$\mathbf{P}(\theta)$

P_{11}

$$\begin{bmatrix} P_{11} & P_{12} & 0 & 0 \\ P_{12} & P_{22} & 0 & 0 \\ 0 & 0 & P_{33} & P_{34} \\ 0 & 0 & -P_{34} & P_{44} \end{bmatrix}$$



random orientation